

Argo CD — A High-Level Primer

GitOps continuous delivery for Kubernetes / OpenShift — the pieces behind your MAS install

What Argo CD is

Argo CD is a **controller that keeps your cluster matching Git**. You describe the desired state in a Git repo; Argo CD continuously compares it to the live cluster and reconciles the difference. This is **GitOps**: Git is the source of truth, Argo CD is the robot that applies it. It runs as the **OpenShift GitOps** operator in the **openshift-gitops** namespace.

The core objects

Object	What it is	In your MAS install
Application	The unit of deployment: one Git source (repo + path + revision) → one cluster destination (namespace). Argo syncs its manifests.	dro.drroc4, manage.drroc4.drgitopsapp, mongodb, operators
AppProject	A guardrail / grouping for Applications: which repos, destinations and resource kinds are allowed. A tenancy + RBAC boundary.	the mas project — all MAS apps belong to it
App-of-apps	An Application whose manifests are <i>other Applications</i> . A root that fans out into many children. Delete/desync the root and children stop being managed.	ibm-mas-account-root → generates suite/manage/sls/dro/...
ApplicationSet	A factory: a template + a generator that stamps out many Applications automatically. Its controller re-creates children if you delete them.	(usually none — MAS uses app-of-apps instead)

If you know OLM operators, here's the mapping

Argo CD reconciles apps the same way OLM reconciles operators — both watch a desired-state object and make the cluster match. The analogy is close enough to build intuition:

OLM (operators) — familiar	Argo CD — equivalent idea
Subscription — 'I want operator X'; OLM installs it	Application — 'I want this Git content'; Argo installs it
OperatorGroup — scopes which namespaces an operator may act in	AppProject — scopes which repos / namespaces an app may use
CSV — the installed operator version/bundle	synced resources — the live objects Argo applied
OLM operator — the reconciling controller	Argo CD controller — the reconciling controller

(Analogy only — they are separate systems. In fact Argo CD *deploys* OLM Subscriptions as part of its manifests, which is why an Argo app can own operators.)

The two things Argo always reports

Signal	Values	Means
Sync	Synced / OutOfSync	Does the cluster match Git? OutOfSync can be harmless drift (an operator edits a field Argo owns).
Health	Healthy / Progressing / Degraded / Missing	Are the deployed workloads actually up? This is the real success signal — not Sync.

Sync policy controls behavior: **automated** (auto-apply Git changes), **selfHeal** (revert manual cluster changes), **prune** (delete resources removed from Git). Your MAS apps run automated — which is why, to uninstall, you must stop auto-sync first or Argo re-creates what you delete.

How deletion works (why teardown order matters)

Deleting an **Application** can either leave or remove its workloads, controlled by a **finalizer** (`resources-finalizer.argocd.argoproj.io`):

With finalizer (cascade)	Without finalizer (non-cascade)
Deleting the app also deletes all resources it deployed. Clean, but can hang on OLM/CR finalizers.	Deleting the app removes only the Application object; workloads stay running for you to remove in order.

Key rule for teardown: delete the generator first — the **app-of-apps root** (and any **ApplicationSet**) — or it recreates the children you just deleted. And Argo cascade does *not* cleanly remove OLM operators (it deletes the Subscription but orphans the CSV), which is why operators are removed Subscription + CSV together.

Quick reference — Argo CD in your MAS install: openshift-gitops (namespace) · mas (AppProject) · ibm-mas-account-root (app-of-apps) · Application per component · AVP sidecar resolves <path:> secrets from AWS at sync time.